

Vicente Raja & Jeff Beck ~ Active Inference Insights

007 ~ The Markov Blanket Trick Debate

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hello everybody and welcome back to the seventh episode of active inference insights I your host darus PV Wayne and today I have for the first time two guests to bring you Vicente Raja is a research fellow at the University of meria and affiliate faculty of The rotman Institute of philosophy at Western University in Canada his research lies at the intersection of philosophy cognitive science neuroscience and the history of the Sciences of the Mind in 2021 he was the first author of the paper the mark of blanket trick on the scope of the free energy principle in Act of inference which argues that quote the Fe is not the general principle it is claimed to be and the act of inference is not the all-encompassing process Theory it is purported to be either we also have Jeff Beck Jeff is a computational neuroscientist studying probabilistic reasoning in humans and animals he is also the senior machine learning research leader versus and is a critical contributor to the active inference field writing among others the the paper Mark of blanket and Mark of blankets and basian territories guys thank you so much uh for joining me it's an absolute pleasure um you know we're talking off camera there may well be some points of Divergence but we are four days away from Christmas at the time of recording and I think the baby Jesus would want us to find some conciliation so we will do we will do the most we can to strike at differences and elucidate differences and see uh but all in the spirit of hopefully collaborative science and uh collaborative thinking so in that with that in mind I think where we could start is actually where we have no issues with one another so looking at that paper where are the kind of points of just neutral classd this is like a good starting point vient I don't know whether you want to start given you wrote the paper but where do you see where Jeff's work or the other key contributors active influence where do you see that you know some of the stuff they say is actually pretty uncontroversial good question um so first of all thank you for for for having me here um um I think I could start with a very brief detour on why I wrote this paper or why we wrote uh this paper on or even why the paper is the way it is uh first of all I wrote I started writing it to actually to make sense of the FP and active inference for

myself right to try to see how it was Drive the steps and actually section two of the paper remains there just as that right as starting with a random dynamical system and just trying to go step by step there so this is I think one of the points in which we converge is at least from my point of view as a person who works in the philosophy of the cognitive science and the cognitive science themselves that dynamical systems is a good way to look at cognitive systems so that's I think we are in kind of the same camp then uh maybe the modeling strategy that people like me that comes from radical embodied science and ecological psychology to model dynamical systems and the Bayesian twist of it that you find in active inference are slightly different right but you know the this first step this first point I think is a point of convergence to um but then after after we we made sense we think of of of of the of the free energy of the free energy principle and active inference ER we our point was okay let's accept all these formalism is good right we don't challenge it we just accept like this is right but even accepting that uh we have some questions about it that how that those questions have to do more on how different bits of the formalism relate to different aspects of cognitive systems and their relationship with the environment right um so and I think this is where the Divergence is stronger so I think the convergence has to do with the Primacy I don't know if the primacy if it is the word but you know the attributing importance to a dynamical systems take on all these issues and the good spirit of trying to find formal methods to study it and our Divergence I think has more to do with uh with some conceptions on how to best address dynamical systems or cognitive systems in dynamical terms and all that stuff yeah excellent okay that's that that's great that's a great start um Jeff let's go to you let's imagine that you're reading the mark of blanket trick for the first time where we will come to the divergences I promise you well I I I can tell a story or I can well so I I'll start in the spirit of this I'll start off with some with with the point of agreement that I had with with the paper which was one of one of the there there there are two sort of often discussed problems with like the FP literature one is that it's very it's very it uses idiosyncratic notation and it can be very difficult to understand um the other one is I is the thing that apparently causes a lot of ey particularly from philosophers and and I agree that this is a bit of a problem which is there's a a penchant for rhetorical excess if I could say such a thing which is a thing I don't think that there's many many problems with the FP as a whole but I think that it could be presented in a more compact and concise way that avoids you know some of the more philosophical claims that are often sprinkled throughout these manuscripts um and I do think that would create both you know and that this is something that

would create both a bit more clarity and it's something that like you know people should poke back to like when you say like if you find yourself and this is my personal view this is not like everyone's everyone who does this if you find yourself putting the word thing in scare quotes you might want to think maybe you're doing something wrong right it's just sort of like my knee-jerk response um and so you know that's that I think is certainly a point where there's definitely going to be some agreement which is is that you know for me mathematician simplify things be explicit that's the beauty of it don't add things to the equations that aren't necessarily there um but of course everyone every scientist does this right we all want to sell our ideas we want to make we want to make them as big as possible and so there is some of that and I think that in particular because the math is so difficult to understand and follow right ha you know has um that those sorts of excesses have become the Focus right that's easier to sort of like say oh that's you know this is a problem this is a problem and so yeah that's that would be the point of agreement if I were to quickly and compactly summarize the point of disagreement um it's I I will do so by telling the following story so when I first came across um Carl FR I was at the Gatsby at the time and um so Carl friston was like literally in the building next door and that's when I when I first came across him when I saw him present his stuff and I read a couple of his papers and and um for me there were many things that I was like concerned about let's just leave it at that let's say concerned we can get into it more later but the one thing that wasn't controversial at all in my opinion was the markof blanket part of the story and that I think and I and my hope is is that you know is that as this conversation proceeds we'll be able to like I'll be able to explain what I mean by that um and maybe come to some sort of consensus so so I'm happy to do that now but I I feel like yeah yeah we we we unpick stuff gradually that's quite that's that's quite a good irony uh that the markco blanket was kind of the point of ah this one this bit works yeah I mean I I take your point that the maths ought to be elegant parsimonious um as much as possible but unfortunately or or fortunately any gr unifying Theory comes with it the philosophy the cognitive science the Neuroscience the theoretical biology and so on um and so I think it's an inevitable consequence that we get wrapped into epistemological and ontological questions but we will do our hardest to unpick them with Simplicity because I think that's actually all that we're aiming for so I read the I read the paper obviously I read it multiple times and I've read the the commentaries and and your response to the commentaries and for me there were about two or three distinct critiques of the free energy principle and active inference that I think we could have a good conversation about the first is that Mark of blankets failed to capture all possible

types of conditional independencies so for example candles candle relationship to the external environment or groups of people cultures societies their relationship to uh the whatever those external Dynamics are that they're shielded from by the markco blanket and then linked to that and then I'll leave it there is that these things to use Jeff's quot marks have properties certain things have properties that cannot be captured by the mark of blanket formalism and viente you mentioned relational properties and constitutive self-organization so I don't know whether you want to take those as two separate things one being um the incapacity of the markco blanket to capture all types of conditional independencies um and then those two strands the constitutive self-organization and the relational properties I don't know whether you want to take them separately or whether you think they fall under the same blanket if you pardon the pun okay we'll do it separately Jeff is just pointed to me let's start therefore with um incapacity as I said for the mark of blanket to to capture all conditional independencies and let's use for sake of Simplicity this uh object that I selected before the candle what what does the candle tell us about the free energy principle viente please um actually we we use candle Flames because it is an example that was in the in the in the literat to right it seems if I remember well I haven't I haven't reread the the paper so I might be not fully accurate about what we claim there but if I remember well the issue with the mark of blanket is is is that what we can take as blanket states are are changing to fasting time so so so there's no way to to find them and to and to know and to and to measure them over time in in in the way that is required by the formalism and if I remember well Thomas par H talks about it in his in his uh commentary to the to the to the marcoo Blan treats it's actually the only commentary that concedes that we might be writing something in so which is an interesting thing but it's okay H and he claims that it could be solved in if if in in this kind of of systems or in this kind of things quotes um um uh we look to instead of looking to States themselves we can look to to to densities of of States you know we can try to model those States and I think this is an this is an interesting move that points out why we think that there are there is more about the marov blanket uh than just to say oh no it's just the simp answer to how to differentiate this you know the internal and the external States right the the the the free energy principle uh and and active inference doesn't really tell you whether you have to choose States or densities of States as the blanket right so you are putting assumption extra assumptions there on how you choose to do that and this is kind of the overall thread of the critique when we claim the free energy principle is not the principle it it claims to be we we are not claiming that it cannot be used as the as the as the principle of Le action to you know to to write down some equations but that it seems like just

from this principle and from answering two simple questions: what is to be a thing and how to differentiate from the environment you have everything and we try to posit examples in which you always need an extra step you always need extra assumptions that are not always clear in the in when you only appeal to the formalism so this is the role of our of how of why we think that the fact that candle flames as an example that was in the literature why candle flames are important I think they point out that you always need you always need extra assumptions that are not always clear in the in when you only appeal to the formalist great um perfect that's a really nice exposition of the claim that you're making uh Jeff I'll jump to you then um it would be you know uh R talk to that would be wonderful and I think actually with specific mention if possible of this dichotomy that viene mentioned which is are we talking about States some kind of ological thing again or we talking about probability densities and where does that line become blurry what actually is that difference or is it just are we just making a kind of false identity claim what what's the sort of what's really going on here um so I think that the best way the best way to respond is would be so the the the short the short piy and dismissive answer is to say I I solved the pro I solved the flame problem I wish I could share my screen I have some really cool like I have some really cool graphics um so what one of the one of the other problems that that that and when conveying the free energy principle that um its its proponents have faced is the fact and I sort of say this the math is really hard and in order to in order to make it tractable a lot of additional assumptions get put in along the way just to make the math easier right and some of those assumptions are necessary to make you know many of those assumptions are not actually part of the free energy principle by itself right or part of the marov blanket story on its own right but are done to make the math easier like when I first came across the FP I thought there was no difference between active inference and KO control theory right and the reason why was because when they go to actually solve the equations they make some additional assumptions to make the math easier so that it ends up being equivalent to KO control theory but if you go in and sort of say well what actually is the principle itself as derived either as Carl does it from um uh statistical physics or the way I prefer to do it which is directly from information Theory right that's not the case right the actual like objective function of active inference and allv solving these like complicated path integrals and things like that but to make the problem easier to make the math easier to make it more solvable additional like factorial assumptions are built in and a lot of people jump on the fact that like oh it's like you have all these additional assumptions they kind of like but those are done often for like computational reasons they're

not necessarily part of the we like to joke that you know you know everybody talks about the generality of the path integral formulation of and and Jan's maxent and how it's mappable on thisal physics what people often forget is the problem with the path interal formulation is you actually have to solve path inters which is a serious pain in the butt and so this is sort of like my summary I what one of the things that has contributed to a lot of confusion has been these additional mathematical simplifications that are then taken to be part of the theory when they are not um in regards to the flame story because I know I gave like a a py and dismissive answer and I'd like I don't you know that was just I will fix that now in the interest of politeness um one the reason that I didn't think that the marov blanket story was like at all controversial was because of its link to systems identification Theory right the Markov blanket itself basically just specifies the domain in which systems identification Theory can apply right so the idea is so what so how does it so when you I don't have you know so system environment inputs outputs right if you're doing systems identification Theory you're basically saying well what is the system doing how what is its input output this is very colloquially right we can make this more what is it's input output relationship in order to do systems identification Theory right you need to know what that input output relation that's the goals to figure out what that input output Rel and if you see a particular input output relationship that gives you the label for that kind of system right it's a system of Type X because it has this input output relationship the markof blanket is basically just specifies the domain in which you are going to do a statistical version of systems identification Theory right it's the domain of the inputs and outputs of so when somebody says I have a big system and there's lots of stuff in it and I'm going to segregate out a little system from it the mere fact that you can do that to when you say like this is a subsystem you have already said that there's a Markov blanket does that make sense right all subsystems have blankets like by definition it's it's it's sort of implic that's and that's why I never thought that it was controversial because what do we do we want say well that's a subsystem that's an important thing it gets its own label and so it has a blanket it's like it's it's it's just it's the way we do systems identification Theory so I don't know I want to hear if I want to see if Vincente has a response before I move on to the flame in particular I mean that that's that's kind of I that's that's that's the that's what I read when I when I was reading the literature more uh it's kind of it's it's it's it's h noncontroversial for the people who work on it in that sense like you you identify a system uh there therefore there's there's there's a blanket and there's an external State I think and part of the Marco blank trick is to see different issues we find with that that that for many systems we are interested in we don't see it

to be so straight forward like we when when we model with dynamical systems for instance with the hkb model H to the the behavior of two couple oscillator where the where the collective variable is relative pH our system is the couple oscillators so you can put a blanket around it I guess but then what's the environment uh it's it's it's the it's the blanket put on the face of space of the relative face which is not a property of any of the two things there right so I mean I kind of understand the abstract sense but then I don't exactly see how this apply to kind of systems that we are interested in right so I I'll just quickly respond so so the problem with the whole Markov blanket part of the story is that there's just two goddamn many markov blankets right you have this big complicated system right and even you know and even if you're going to limit yourself as they often do in the FP literature to like static Markov blankets right where one once a particle or constituent component or whatever has been assigned to either environment blanket or object it's always there right so this is a situation where there's no such thing as matter transfer right which is why the flame Case by the way like doesn't work right the flame the Flames have Markov blankets but they're not static right because as the matter you know as it burns and moves in and fuel comes in and and exhaust goes out right the matter is constantly so if if your definition of a blanket is that it's tight to matter then yeah you're right like you can't do a flame at all but that's so but putting that issue aside the problem with Mark blank is is that they're everywhere you have a glass of water every single fluid element in the glass of water so what's a fluid element right you take a glass of water you draw a little circle I mean obviously it's not it's 3D right but but an arbitrary fluid volume has a blanket right and we and and we know this like when we do finite element methods for simulating them we draw the we take our glass of water and we make it into lots of little cubes and all the cubes are connected and the boundaries between the cubes form the blanket there's too many blankets blankets are absolutely everywhere right and they and their boundaries are often completely like where you're going to put your blanket is the hardest problem to solve right and I think that that that the free energy literature has not worried too much about the problem of where you're drawing the blanket because the idea was if you've got a blanket then it's subsystems have blankets right we're interested in subsystems and so now we can just like put that problem aside but it's a really difficult problem to say well what is the principle that tells me that this gets a blanket right why why is a blanket that's around you know a bubble Rising through my glass of water the right blanket as opposed to an arbitrary blanket around a fluid element that has no bubbles in it right and I don't think that that that question has not been adequately addressed in the you know you know in my

opinion but I believe that there is a notion that is sort of implicit in the Fe approach that handles that and so when when I when I sat down and tried to solve the flame problem I asked myself well what additional principle and this might make vincenta happy right there is an additional principle he was absolutely right I think the principle was implicit um but the additional principle is like how do you choose your blankets well you choose them for parsimony and simplicity right in other words we use the scientific method right you take all the ways that you you have this complicated interacting system you take all the ways that you could draw the you know boundaries in them and there's like an uncount you know not uncountable but there's a really large number of different ways that you could draw your blankets and the way you pick the one that's important is the one that results in the greatest simplification of the of the resulting Dynamics this is basically like mean you can think of this as being field or course graining if you're a physicist right that's what course graining is that you segregate your system into macroscopic variables or macroscopic components if it's a mechanical system and the choice that you make as to how you're going to do that is to is to basically make the system as simple as possible if you take that approach then you have if you take that approach and apply it to the flame situation in particular that's exactly what you discover right you discover that the blanket of a flame right is just outside the reaction Zone where all of the exothermic reactions are going on okay excellent no I think we're agreeing too much um I think that's going very well um I love the the flame problem I like that yeah gorgeous I I really wish I could share my screen because I like queued up some videos and like you know and had some figures that I was like just in case the flame case really is cool I'm gonna publish that soon I'm just working on I'm trying to get like a whole I'm working on a um I took some fake I took some data from you guys know what particle lenia well you know everyone knows what Conway's Game of Life Is So This is Conway's Game of Life but for interacting particles and so they get these particles to do cool things and so you can put in a random bunch of particles and they form a cell membrane and the cell membrane grows stuff like fella and so we applied it to that situation as well and what you find is that the blankets are changing over time and as they change the identity of the object also changes which is exactly what you want right and the reason for this is because the blanket itself the blanket alone is meaningless right it just specifies the domain in which you're going to do your system identification or dimensionality reduction right what really defines an object's type is the statistics of the domain right and so as a result you can watch these cells evolve and as the statist as the statistics of the various boundaries in the system because there's more than one

um change so does the identity of the associated object right which is really what want this is really all about labeling stuff right right and in order to label stuff by type you need a blanket and the way that you identify a type is by the statistics of the blanket okay this is really useful um because yeah I think we found agreement with the idea that the Fe needs to integrate some historicity some Dynamics and perhaps even changing identities of you as you've just outlined there Jeff I think yeah well I think the the critical thing is is recogn and this is implicit in everything that Carl writes right it's it just needs to be made more explicit right he says the blanket defines the phenotype that is a misleading statement it's true but it's misleading right the correct statement is the statistics of the blanket Define the phenotype right right it's not just the blanket itself that the blanket itself is just a domain yeah right it's just where yeah so yeah well I think this is a um this problem I think has started to be remedied within the active inference literature um because I don't know if you guys have noticed this as well but there's been a move to turn the internal generative model to a generative model that actually both encompasses across the mark of blanket the external and the internal States so this is something that Maxwell Ram makes very clear in his um in this year's paper um physics of belief to put it roughly um but I think that there is there is still a slight point of difference here and I wonder whether it's just a semantic one Jeff you still talk about internal States or the particle the thing the system and the environment whereas visente is talking about coupled oscillators and the way that I think about this is we can either have an intrinsic or an extrinsic relation between let's say external Dynamics and internal Dynamics now what does that mean well an external relation would be something like a causal relationship so the A and B retain their identity despite the fact that they're causally related an intrinsic relation is something more like a familiar relation okay if I am uh my you know my mother is defined as my mother because of the relationship that she has with me so in many in many ways I find that this distinction that you guys are carving here might be neatly mapped on to that distinction benter I'm going to start with you do you think that a mistake that the fvp makes might be to try and carve out an external relation so we've got a enacting changing B which changes a when really the focus should be on the Dynamics of the entire system as a whole in the way that sort of Evan Thompson and um those who followed spoke about these kind of couple Dynamic systems where the identity is in the mutual coupling good question I mean I would like to to just point out I think Jeff has pointed uh has pointed out uh what our issue is um and and I agree with with pretty much everything he he has said in that in that regard right our issue basically was that I mean we have nothing against marov blanket blankets as a as a

methodological tool as a mathematical tool but we would like to know more about how you select the blanket because we think that depending on how you select the blanket you are interested in you be doing two different things right and and and that and that it's there are it's other parts other fundamental parts of your theory what makes you to to to select a blanket let's say for the organism uh and then the organism has internal States and the environment is the external States or if you are an ecological psychologist like me you want a blanket for the organism environment system and then everything is internal State and it's not clear what the what the external state would be there right you can also choose a blanket on the organism environment system and a nested one and another blanket it's it's blankets all the way down man all the way to protons they got blankets too so as as a mathematical tool so to speak there are too many blankets right yeah but how you choose them reflects what kind of what kind of fundamental Theory you have in mind and when I say fundamental Theory I think about psychology cognitive science right so um if you for instance think that the organism has this input output mapping and all the relevant computations happens inside so all the control of behavior all all the are all the all the all the perceptual inference is depending on the internal States you you are first you are likely a cognitive scientist of the mainstream right and you want the the blanket in the organism and the external states are basically input right uh while if you are an ecological psychologist and you think that to understand the perception for instance the first step is to see the structures in the environment those structures are relevant in themselves and contribute to perception as much as the internal states of the organism there so then you probably don't take those structures to be just input right so my my point being here that I totally agree that there are too many blankets and that this selection process depends on deeper theories that go maybe beyond the free energy Principle as such and you can then you can use the free energy Principle as a method to to develop those those theories right yeah so yeah so this is the I I mean my take on this is so I I completely agree right there there is a sort of hidden additional principle but I think that one of the reasons this got neglected in the free energy literature is because the whole point was we're always drawing blankets you can draw it anywhere and this Theory still applies right doesn't matter where you put your blanket right I can still I can still treat the system as if it is doing inference over its environment it does you know exact you know that that that was sort of like I think the reason why the the sort of additional like how do you select your blanket problem was neglected was because they were trying to show the principle always works right um that's sort of my my my defense of of of of of Carl friston's world is that like there's a reason they skipped the problem because wasn't the

important problem right the important problem was it doesn't matter how you draw your blankets you still get you still can apply the rest of the principle so if you decide that that your system is a cell that's fine if it's a collection of cells that still works if it's it and so on right so our question was then I I think this question was in the Marco blancet trick too what if your blanket is the is the is the organism environment system what are the external states to that system that the system INF first so to speak if yeah if if you're if you're yeah so if your blanket is like the whole universe I don't I'm gonna go with it's Christmas I'm gonna go with baby Jesus okay very good right no right I mean you know the so I think that that's counter to the to the reason why we want to Define blanket in the first place and the reason is because we want to label things right and the reason we want to label things is you know is because well it it simplifies our lives right if I can look and say that's a cat that's another kind of cat right then not you know I can sort of use you know that you know use those labels because they make my my life easier right they make prediction easier about what's going to happen and things like that right that's the reason why we want to have labels and the mark s well you know this is a potential place so that and the reason I make this because I do think that implicit in the whole approach while not necessarily as explicitly made in the literature as it should have been is that the goal is to label things the goal is to come up with phenotypes and since the goal is to come up with phenotypes you can ask the question well why would we want to do that and the answer is parsimony right we want a simple compact representation of our world and the way we do it is by taking big complicated things made of lots of little parts and giving them a single label that describes a lot of their General properties and features and that is exactly what the mark out blanket does or is capable of doing and so that also tells you which is why like in my world right that's the additional PR it was implicit I'm I want to make it explicit that the way that we choose which blankets to focus on is like the way we do science it's based on simplifying Dynamics for the purposes of prediction and data compression Jeff I may jump in here um and perhaps speak on vincente's behalf I'm very happy for him to jump over me and and and clarify this but I think a question comes up here which is so we're interested in labeling we're interested in describing coupled systems so how come when in the FP literature we draw blankets they so uh coincidentally happen to fall elegantly on what we intuitively believe to be things and their blankets so for example yes you say okay it's because well we see cats and glasses and humans and planets and like why wouldn't we carve blankets around them but why in a sense does it work to carve blankets around them if we're dealing more in information exchange than how the object appears to our sensorium and given that as

a mathematician or a physicist what does that say about the mapping between kind of our perception and the information transfer that we're being able to describe via the Marco I think very hard about that question um because my my knee-jerk response is to like is to launch into my like what is science for discussion which is probably not going to directly address uh what it is that you what what it is that you want me to to address um when when the issue is still if the issue is still like um let's see so there are two there are two things going on here right one is what do we do with the information that the Universe gives us how do we categorize it label it and make it simple and easy to understand and then the other issue is is well what is NE you know what is any subsystem necessarily doing right and those are very different issues right when we talk about the when we ask the question though why do we call this a subsystem and and you know and then like I'm trying to cut it in half and like this not and and this not a subsystem and that choice is arbitrary right that's one of the the lessons that I'm trying to drive home here is is that like you can draw a blanket around anything the question is why would you right having drawn a blanket all of the free energy mechan the basian mechanics all apply right but why why would I draw a particular blanket and it really is because it's useful to us right if we could just like have a computer that simulated absolutely everything right and that didn't and it was like I don't know it cost like five cents to run we would totally do that and we would be happy and we wouldn't care about labels right but we don't have that ability right we are trying to construct these you know things that we can wrap our brains around we're trying to simplify things you know to make them parsable and understandable to us and so that's why we would choose a particular set of blankets now but again the point though is and the reason why I think that there wasn't enough emphasis in the free energy literature is that all of once you have established that there's a blanket which you can always do right the rest of it just follows right all the stuff about you know is it effectively doing is it behaving as if it's performing inference on its environment yes right I think the more interesting question that Vincente brings up in the paper which which which I read yesterday that I thought was a really great question um was had to do with with well which is the environment and which is the object right because even this parsimony principle that I just established right doesn't help with that because I just said simplify the global Dynamics like what what you know what what um macroscopic Theory does the best that happens to have this partitioning um and that doesn't tell you which is the but this is but this is where I think you now need to move back into physics land right so people like I I really like the work of Jeremy England um and uh and everyone loves like shinger little I is it what is life is that the title of it right um where there is

you know in in in that work there is like a a definition of how you there is a notion of how you tell which is the agent and which or which is the object and which is the environment which is the agent and which is the environment and it basically has to do with energy flow right so you could sort of say oh well you know the organism is the thing that takes in energy and produces entropy right and that once you've done that if you're talking about a physical system right you know then now you have something that that does in fact uniquely identify you know which is which which you don't get out of then this is my point right you don't get that out of a out of a Simplicity parsimony type principle right I'm not sure whether you wanna yes go that second point or whether you want to loop back to my first point about the mapping between the supposedly arbitrary epist logical tool based approach of the Marco blanket and actually maybe how that maps onto onto logically at least what we perceive I don't know which one you want to start with yeah I think that that that the the last uh thing or did this last in intervention by by by je I think it he States and I think we we are going to agree here he he literally States the Marco blanket trick right that that as as as soon as you put the blanket on top of a system right or whatever you choose with all this uh pragmatic stuff then you can do Bas inference right you as you put the blanket you have now the so to speak the data structure with latent variables observables and internal ones to do that and I think we agree that or we could agree that this is the Marco blanket trick while Jeff thinks this is good right uh it's it's it's it's good we think it's so so right but that kind kind kind of be the point right so I think the Marco blanket trick is good if you are already convinced for instance that uh perception is a kind of inference and then it gives you very precise mathematical tools to model that now if you think like some crazy people like me right that perception is not something that you know something that the internal states of an organ the internal State states of an organism do in order to know the external States so to speak but is a property of the organism environment system then this data structure doesn't really map very well with what we want to do or maybe it does but we don't see how and and and part and part of of our paper is to is to try to to express you know to to state these doubts on okay how could apply this if I think this s thing right and uh and then second part uh which is what the what the um why the blanket falls exactly where it needs to to be I think this is a part of first this is a part of a conversation I've had like dozens of time with Maxwell right he he's he's the person I've talked more about this so gve talk about this and actually in in their commentary to the Marco blanket trick they talk about this uh paper he has with dton the weak marov blankets right that they prove that there are a lot of them right like like as soon as you have Dimensions you know and and our

question in our response was yes I mean but the point is not that you have a lot of them it's how you choose the right one so how you choose for let's say for the visual system how you choose the blanket in the sensory receptors so to speak right and and I think Jeff has been touching upon that right in in in in some of some of the things he he has been saying today before I move back to Jeff could you just maybe for the audience expound upon what you meant by this sort of Distinction um Mak sense when we take for granted perception as inference I think that would be quite an interesting Avenue just to unpick a little bit further just so audience knows exactly what you mean by that yeah so uh long story short uh the common theory of perception at least since handhold but most likely seem the cart or even beyond that is that ER perception is is a is a form of inference right so the our mind brain is in the business of inferring the real environment so to speak from impoverished stimulation so what we get in in in a our sensory systems is usually ER ambiguous not not not very informative or not completely informative and generally uh insufficient as to know what is the exact cost of the of that input right like the typical examp ex is is that is that you know a a a small close object can give you the same input that than a larger object that is farther away right they can occupy the same angle in your in your retina blah blah so the mainstream theory is that the brain is in the business of inferring the real properties of the environment from this impoverished input and it that's so because it it can use uh prior knowledge memory experience you know to build up and good model of the real environment ecological psychologists think ER along with many other people that come that come at least from Willian James Willian James explicitly claims that the kind of um of approach this inferential approach to perception is pure mythology I I mean it's kind of the way they wrote in the 19th century right but the point being is that for ecological psychologists the kind of prior knowledge you need in order for the inference to work is very difficult to justify um so they think that perception is more of a relation of Rich information that comes from the environment and the and the internal Dynamics and the relation itself is what uh perception is but we don't need an inference because we don't need to go from impoverished stimulation to the real environment but the the there's a way to understand perceptual information that gives you all you need without the need of prior knowledge in order to get grips so to speak with the Real Environment so it's kind of a direct undirect staff or perception as an inference or perception as an as information detection now I have to put again because I've talked with Maxwell a lot about this so I just want to put a uh so sort of um here right you can still see how the internal Dynamics change in order to couple with these and reach information in terms of some form of basian inference

because Bayesian inference is yes you know you can describe the the path of a dynamical system towards an attractor in terms of a probability density of the state so there's there's a form to to maybe there's a form to use inference that is the Bayesian inference way in which you capture both of them I'm still in an ongoing discussion with MaxW about yeah this is this is the as if part of the free energy principle right the math tells us that everything right behaves as if it is performing Bayesian inference right it doesn't tell us two additional things like how shitty that inference is it could be real like a rock is performing Bayesian inference it's just not doing a very good job um and uh you know but it but that's but that's that that just follows that just falls entirely from the mathematics up right is that is that it is an as if description right the description of anything you know it is always an available description of a phenomenon that it is that it functions as if it's performing inference right um whether or not it's actually performing inference or doing like solving the right inference problem or or however you want to put that um is not explicitly addressed and I think a really interesting question it you know so for example when you know when one applies the the free energy principle in practice there's always this implicit assumption that whatever the thing is doing is it's actually doing inference correctly right there's an optimal inference engine sort of like hidden inside now they can get away with this right because what goes on on the inside doesn't matter right and what defines the object is the what defines the thing is the statistics of its boundary and so if you have like a really boring boundary statistics like you're a rock right then you don't need a very sophisticated like like the inference engine that you would actually use to sort of say well what is the you know what does this rock have to infer about the external world in order to be a rock and the answer is very little so you kind of get a lot of these things for free but it is an as if description right it's not like you know it's it's just and you know but like I this is you know another way to get to the free energy principles from a purely information theoretic approach right and I prefer that approach because all of this stuff makes a little bit more sense right you're not sort of saying like the rock is doing inference you're saying you know the rock is behaving as it becomes more clear that this is sort of an as if argument right as opposed to it's there's literally an internal state in the rock that is computing you know the probability that the that the water in the stream is about to change no it's not right okay yeah I I had only because have to in order to be a rock yeah yeah yeah I um I was did a quite a good podcast with Adam Saffron a couple of days ago and I said my favorite two words in the free energy principle are looks like um but as as if is that as if as if is the one that I like to pick up yeah as if looks like uh no I think that's I think that's useful clarification um unless you're a pan

psychist and then they might be coming for you um what's wrong with pan psychism I don't understand why everyone's down on hey I I actually don't mind that I'm kind of into the spookier versions but well they might they might be Keen to perhaps I can't speak for Philip Goff and everyone else but they might be Keen to bring into question the validity of that as if or looks like but we need not go there um I wanted to sort of I do want to this was one of the critiques that I kind of identified so I do want to just get some clarity on it um and I'll actually quote Vicente here so this is vicente's response to the commentary um the extent to which something counts as a thing depends on the choice of Market Mark of blanket partition which can be adjusted according to the needs of the model there is an ambiguity here on the one hand it is claimed that things are what they are by definition even if boundaries do not real align with the intuitive ones for ordinary things on the other hand when the Fe presents possible examples of things like brains or cells the mark of blanket always falls in the right place from the intuitive point of view so that probably puts more elegantly the point I was trying to make earlier je does that shed any Clarity on where I was going and Jeff do you think you do you have a kind of repul to that I I almost hate to do what I'm about to do can you can you give me the quote one more time my bra I'll give you I'll give you the that was was a lengthy quote and I'm gonna need to hear it one more time to fully absorb and appreciate it I'll do a sort of amended version a shortened version on the one hand it is claimed that things are what they are by definition even if the boundaries do not really align with the intuitive ones for ordinary things on the other hand when the FP presents possible examples of things like brains or cells the mark of blanket always falls in the right place from the intuitive point of view okay so I do have a response to that and that this goes back to the notion that what defines a thing where you get this sort of notion of phenotype is the statistics of the blanket and so that gives you the ability to sort of say oh this is like you know like right now I'm I'm I'm concerned with brains so I all I have to do is ask what is the statistics you know what is these statistics of the of the brain blanket right and then that's the thing I'm currently going to investigate right and so you know what but I do think that this does illustrate the you know one of the issues which we've been talking about which is that like there is this sort of fundament you know if you were going to like denovo start just putting blankets on everything in the world right there isn't an explicit principle present in the FP that tells you how to do that right there's this additional principle which is like you know that that that I that that led to the you know the algorithm that I used to discover the blankets of flames which is like parsimony right but it's it's not now now the reason why that that they don't worry about that is because

it's the implicit in every single thing in the FP literatures notice that we are labeling stuff right I have the freedom to label what I'm interested in and having done that right that sort of tells me what the blanket statistics are going to be um I'll just add one other thing the reason I prefer like the the information theoretic derivation of the Fe is because is is because um the way that works is you start out with a probability distribution over blankets right it's like this is like is and you take it's a it's a it's a ma it's a James Max cow or Max ENT kind of approach and you sort of say like hey like I have some laws that tell me how the universe works and those are some constraints right and then I also have I want to find I want to find some subset of my system that has these statistics those statistics are then definitional this leads to you know sort of Levan equation type things right that you can then solve which tells you how the world has to be how the individual particles in the system have to behave if you're doing it in particle space um in order for that thing to exist right and it has to exist because you imposed it as a constraint when you did your sort of Max entty thing right um if you have it if you have that in mind what you're really asking when you sort of when you take this sort of Fe you know when you we sort of we if you take this sort of um can I can I use the phrase ontological potential or is that g to make people upset right one of the things that comes out if you if you take this sort of free approach you're basically like look these statistics Define an object type right and now what I want to know is I want to know like well what what's inside what's outside like what kind of environment can it live in like what kind of comp you know you know what what does it have to be made of right you can just apply that those statistics as part of a constraint do Max ENT and get all of the equations that tell you how what kind of environment this sort of thing can live in right but the starting point was like I'm interested in like this kind of thing right and the reason why we call it an ontological potential is because you sort of said like you've You' started by saying this is my definition of something and free energy is literally a potential function like when you do it in this Max nway it's literally a potential function so free energy is an ontological potential but it does start by saying this is the thing this is the thing I'm interested in it starts by specifying those statistics the statistics of a given object type do you want to add anything to that I mean I'm wondering whether from a philosophical if you're putting a philosophical hat on what prespos positions just that kind of as as Jeff said like we're starting from like this is a thing what philosophical presuppositions does that uh entail yeah I mean uh I'm I'm kind of happy with that right you choose your thing and you you you apply the methods you want to so I'm we we we we really don't have anything with when I see with the authors of of of of the paper of course but myself also uh we really don't have any any uh

criticism to that our uh criticism and actually it was that paper was published along with a few more that were talking about that there's this this this one by J Brun Cal on their blanket and Miguel Aguilera and Chris Bley had another one there right so so to speak if you are a front with look this is what I took to be what I take to be a thing or this is my thing of of interest for these reasons and then I'm I'm going to apply the Marco blanket here if the methods work go you know like cool uh the thing is that sometimes in the literature these extra steps or these parts are as Jeff has just said not explicit enough so it seems that everything follows from you ask what is to be a thing and then you have a full theory of cognitive neuroscience and we are like we don't think so right there are other other other deep ideas that word there right and but if we go back to the to the to that quote um it's also not about the kind of the the the general possibilities of the inference and the Marco blanket formalism but more about how it is presented in the literature sometimes right so sometimes it is clear that you put the blanket so in so to speak in the in the space not in the real space but in you know you you you select some states and then you put the blanket there and this is clear for instance in one paper I think it's Michael Kirchhoff Julian Kerstein Carl Fron and I don't know if I and if I if I forget someone else but they actually modeled couple the pendulums right and the blanket states there where something I'm not completely sure but where something like velocity and yeah position or position acceleration these are not the the intuitive ones right this is not the the the surface of of of of the pendulum so to speak but then when you go to any figure in the in the in in in the literature you see the blanket that is in the in the sensory receptors or or or or or in the membrane of of of of the cell when you expect the blanket would will be from an intuitive point of view so our point is that the these two claims seems to be not very well aligned right either you find the blanket in an in an abstract state of space or or or or in an space of states or or or or or in an N dimensional states of space and then the blanket can be velocity acceleration jerk whatever you want right or you go to the more intuitive uh way in which you know the the blanket of the cat is you know I guess the hair or something like that or you find a way which which I guess is the is the is the is the ultimate aim right you find a way to align these two maybe but I I think the reason that this happens though is because because the principle is uniformly applicable to all systems and you know by and all sub systems and all subsystems of subsystems they kind of like glossed over that problem it's like doesn't matter what system you're interested in this Theory applies right and I think that's one of the reasons why like the issue that Vincente brings up was basically glossed over right it's because it was it wasn't you know the the the the two interesting questions from my perspective that got

glossed over are how do blankets form in the first you know how do the blankets that we would like to call blankets because they're blankets everywhere but how are the interesting how do interesting blankets form right what are the what are the mechanics that that lead to that process and then why do we say this is an interesting blanket then that is just a drop of or just a fluid volume blanking on the easy stuff now why why yeah why is that just a fluid why is that an uninteresting blanket and this is where the whole like parsimony thing comes in right interesting blankets have interesting properties right and they simplify the overall Dynamics but that but but if you if you if you start out your paper by writing that like oh well I'm gonna have this other principle and it's gonna say this is a blanket and this is the preferred blanket right then you sort of if it it kind of tarnishes your ability to say that your theory is uniformly applicable to all systems and subsystems and I think that that is ultimately the cause of this of of this particular confusion you know which is why like you know know like when I saw this stuff my knee-jerk response was I why are they saying a flame doesn't have a blanket right so my I I I should mention my my uh the title of my PhD thesis was like what was it complex Dynamics and solid flame micro structure oh actually burning stuff and and the way that we model Flames is explicitly with a blanket the blankets you know is is is there the flame it's burning this way down a fuse like this I think Acme bomb right it's going like this it's going down a fuse and there's a blanket right the blanket is basically just outside the reaction zone right in the same way that like you know fluid flowing or air going by like an airplane wing forms a little boundary which we call curiously enough the boundary layer right um and you know and that's a blanket too we are constantly using blankets because they simplify Dynamics and that was exactly how I got to this you know that's exactly why to me that there was this implicit parsimony principle that solved that particular problem excellent um but but I but I certainly agree like you know it was it was a rhetorical choice right you want to show that it it always applies so of course I'm just going to randomly draw blankets as you know as need be because it always works I I like that you know in the in the in the I mean in the in the spirit of generality they kind of glosses over these tiny details that there are some systems that might be tricky right but I think there's kind of another reason for that I don't think it's a it's a it's an opposite one it's a it's a complimentary one but and and it's a reason that I think applies um mostly to Psychology and Neuroscience um like like Jeff I don't I'm not a super fan of the of what what they call the high road to to the to the free principle yeah the the random dynamical system plus I like uh better the one that comes from predictive processing and builds up right first uh because I think it is it it what happened historically right with the

with the with the 1990s paper on how to encoders to you know and you can see that and then uh because it makes explicit how the free energy principle within Psychology and Neuroscience grows within a theory within an overall theory in which this separation is already taken for granted right you see there are the internal states of of of which are the brain that are making the inference there are some blanket States right which are receptors and there's some environment there so right so right this kind of blanket formalism perfectly fit the overall idea that there we have internal states that have to infer the external one so it was not an issue for kind of say the mainstream psychology Neuroscience to think in that way it was like taking for for for for granted so to speak but then this there's this weird Weir weird people call ecological psychologist that have issues with that thing and and and of course that's from that's the place from from where our paper comes right we see issues in that in some systems in that in that so so so I'm gonna propose like The Happy The Happy Middle Ground here right so so the statement that all things behave as if they are performing inference is kind of hard to get around right because like it it literally follows from I mean it's it it follows from whether you take statistical physics or information Theory it's sort of a statement about like an but what what it doesn't tell you is the thing that is is what it doesn't you know so now you say well I don't think the brain is doing inference or I don't think perception is inference or something like that it turns out that's not incompatible with with with what the free energy principle states either and the reason for this is because the free energy principle says that all things behave as if they are performing inference it says nothing about the quality of that inference it could be really shitty inference it has to be good enough so that the thing can maintain its boundary statistics the the statistics that Define it as like that type of a thing but it could in fact be really bad compared to what a normative model of inference would be would be performing and I think this step this step has also been been sort or this this little bit has also sort of been been missed right there there in a lot of applications because like especially like what I do nowadays which is like you know AI applications you don't want to find systems that do shitty inference right you want to find systems that do optimal imprints right and so but but but in practice right that that becomes the new question sort of like well okay so they've shown that it's always doing inference and I'll just accept that but I don't care I don't care because I'm interested in the exact in in the kinds of inference mistakes that it's making right it's kind of like saying like when we're in the lab right so we I've worked with a lot of Behavioral scientists over the years and when they're in the lab they always sit the monkey down and okay let's not do monkeys uh there's a mouse and the mouse is doing some task and

we always want to ask questions like well how well is the mouse doing this sort of like this task now we can solve that problem because we designed the task and we set the mouse to solve it right and so the um we can sort of say oh well he's doing he's getting like 95% correct or let's say he's getting doing 90% correct H but the maximum he could given the task difficulty the best he could possibly do is like you know n 98% or 96% correct like that so that's comparing the animals Behavior to Optimal behavior and we do that a lot in the lab but when it comes to really complicated things we don't H we don't have that we lose that ability right and this is where cognitive psychology comes in right cognitive psychology is like okay there's no way in hell a human is going to solve this really complicated social dynamics problem optimally in fact we don't even know how to write down the solution to this really complicated social dynamics problem properly now you might be able to say from this free energy perspective that the person is behaving as if they are solving it probabilistically or as if they are doing inference but it's not optimal inference and so now what what becomes the interesting area of Investigation the inter area of Investigation is what are the mistakes that some someone is making how do we characterize those like how do we just characterize their behavior right we don't have to you know if you're a psychologist you don't have to commit to this as inference problem right because the the question that you want to answer is not is it doing inference but it's what sort of you know what sort of mistakes are you making what sort of assumptions are you making not which is a statement about the quality of your inference algorithm which is much more interesting than whether or not you're doing inference I that was way more long-winded so I apologize I meant that to be Compact and concise just didn't it's a it's a really no it's a really useful Edition um so backtracking 15 minutes just for the audience um the paper that viente mentioned is I believe uh kof and colleagues 2018 the mark of blankets Mark of blankets of life I like just clarifying that in case people want to backtrack and um I believe that's the that's the blankets of blankets one right yeah that's the one with the pendulums clarify that okay this is I mean again we're getting on far too well um but no we're doing well we're doing well no no one shouting at each other yet um okay no this is this is super so so as you said Jeff um we kind of need to do approximate Bays sufficiently well that we don't dissipate into the heat bu or it looks like we're doing that um and that's I I I feel like some of the ways that issue got skirted is that we would encode a higher order uh if we're talking about humans a higher order prior preference that we believe that we should be or we are minimizing variational free energy so there's yeah no what yeah is that like kind of uh go ahead go ahead I'm just curious about we we don't have to believe anything about this right

organ a cell doesn't have to believe in the free energy principle the point is is that it's a mathematical inevitability right it's it you know you know so you know would so I guess okay so let me let me go back there are two ways to think about the free energy principle and one of them has to do with like like self-organization so what does self-organization mean sort of like it sounds as if you're sort of saying that like oh I have a notion of self right that that I act so as so that that I act that you know I take action so that my notion of self is coherent I don't like any of that stuff um I know that a lot of other people do but I like because because the whole point is is that you don't even need a notion of self right we it you know in order to behave as if you have a notion of self right the the way it's constructed it's sort of like oh okay it has a boundary the boundary has these statistics what simply has to happen to maintain those statistics those statistics give us a notion of object type which you could call self and you could sort of say well okay for now you know then you could sort of ask the question well inside does it also have this notion of self I don't think it's a meaningful question right it's you know it's yeah it's sort of like well we've we've said like this no you know this is the object type this is its definition and we're really asking a question how do internal and external States collaborate in order to maintain the boundary and if you take the if you take the the max entropy you know uh version of the derivation that becomes explicit what you're really asking is how do internal and external States collaborate to maintain boundary statistic has nothing there's no notion of of like self yes I no I'm completely on board with that I do uh some of the work that I like to focus on is actually about self-modeling self- construction um which completely is opposed to the cartesian ego picture and I think actually a lot of the conceptual problems come from a retention of the cartesian ego that people just feel very it's very difficult to abandon that picture no I I was only adding this because well what you said is very much in line with what Carl said when he came on which is that there is no real existential imperative it's just if you can see object X at time here and object X at the same supposed object at t plus one they've had to have done something that looks like uh Asian inference so it's a it's an exponential imperative only in the sense that we can make the we can we can kind of say that things persist through time but there are these sort of added bells and whistles right so I you know I'm pretty sure there are papers now maybe these are earlier papers which say something like there is I've read it today there's an a priori belief now whether this is a propositional or subpersonal belief I think is somewhat trivial that I am and again I'm saying this to make sense I am the type of thing that will minimize variational free energy so given that but given your uh point that we can do quite shitty dis uh variational basing inference perceptual inference for

one and AR yeah relative yeah relative to Optimal but yeah right AR divergences can be quite big um what Vicente do you see a way of um accounting for the Persistence of the organism Beyond just like this slow optimization problem um which as I said kind of historically at least invoked this high order prior that I am the type of thing that will um do desense down free energy gradients way well I mean I guess this is a question on how do things persist in time or how what we call things persist in time so one thing is that I don't have any issue with the with the mathematical imperative part of it right that's part of of I mean I can say something like given this and this constraints the uh dynamical system with an with an attractor will tend to that to that attractor and that's a mathematical imperative to right if you have an attractor with the constraint may be the system will tell that and actually you can redescribe that tending towards the the the attractor in terms of basan inference in terms of self self-information minimization whatever or maximization and free energy right so I don't have anything about um the thing is is is that the way uh the permanence of things or the self-organization of things without the belief of it has been uh studied in time uh is is the kind of thing that uh touches on homeostasis allostasis or even autopoesis right an activist think that the thing what the thing does is to is to self not not only to self organize but to self uh produce itself right the the the in so to speak the internal state of the systems are engaged in maintaining the boundary right that's what they what they what they do there and I don't think they they they picture that in terms of having to believe that they basically picture that in terms of catalytic or even physiological ways to maintain that right um and and I guess that's that's that's an account of homeostasis too of allostasis when you include the dynamical relationship with the with the with the environment right so now I guess the question is that to what extent or the question we at least we were interested in is to what extent um this physical imperative of the minimization of the free energy or the or the Performing as if making basan inference is able to capture autopois for instance right and not only autopolis not only the self-organization part but the self- production part um and we and that's another question we have in the yeah so so I have two I have two answers right uh the first is is that when it comes to like maintenance maintenance is baked in right if you if you say my definition of a thing is that it has these boundary statistics and their stationary statistics it's baked in right you know which is totally fine that's your definition of that object type it is stable right and that you know which is great the the the hard question and the question that you know that has been I think un that has gone largely unaddressed in the F literature is how do these things form in the first place and this is right like you know when you say like oh it's a um you know the focus has always been on

homeostasis which has nothing to do with starting from a glob goop right and getting something that is stable right it doesn't have anything to do now there's self-repairs in there right and there is like finding you know H days and balance with your environment but there's nothing that is how do I get from you know a chemical soup to a cell that is the F has has historically you know been completely agnostic on that topic right um and that's one of the reason why I don't use words like you know alost stasis I also forget what they mean because that's a big word yeah really big word um but uh but but but but which but but that said there are in fact people working on that particular problem that's why I like to like like so I've mentioned Jeremy England like twice now this he get he get a third mention that's how cool I think is I also just read his book which was which was a lot of fun um and that's probably why it's on my mind so the approach that he's taking is well okay so that's the goal right the goal is to say well what you know what actually you know why is it that these that these complicated things form apparently out of nothing and the math that he uses is is in fact the mathematics of statistical physics and the thing that he and you know and the mathematical object that seems to motivate this is constructing free energy minimizing flows right great he even has a love I think it's chapter seven in his book where he then goes on to say and by the way you could think of this you know these free energy energy minimizing flows as doing a kind of computation about their environment but he doesn't actually say as if I actually read through it twice just to see if he used the phrase as if and it turns out he doesn't which is unfortunate but he makes a very similar argument right is is that is it you know this free energy minimization principle can in fact also lead to the formation of these things the math is a lot more complicated right and certainly static blankets don't apply in this scenario like the STA a static Mar blanket doesn't apply right um because the whole point is that over time it's actually forming that blanket is constantly changing in morphing into a much more interesting set of statistical that that so the statistics of the blanket are no longer stationary so a stationary blanket doesn't um but the same free energy principle applies the same free energy principle that shorting called negative entropy um uh or that was sometimes referred to as negative entropy it's it's and and the reason for this is because it all fall out ultimately from like statistical physics okay fantastic well but yeah but but you're right that has not really been I I just wanted to reiterate I did have a point when I started the tirade I just want to get back to it um you know it is true that that hasn't been addressed like how these blankets actually form in the first place is the free energy principle in its current form has been largely agnostic and that's why we say well take a look what jerem England's doing it's basically for energy too

okay I'd like to get to that maybe as our sort of closing uh remark because don't think there's going to be too much antagonism or or Discord there I will raise one more criticism that was um mentioned in Ezekiel depalo's commentaries so this is his criticism of the FP and and and then we can move on to the fluffy philosophy or statistical mechanics whatever it is um so aalo has two critiques in a way of um this uh of the Fe formulation the first is exactly what we've spoken about which is that we have to make a choice about where to carve up the Marco blanket and we've covered that until the cows come home although I'd be always happy to return to that but his second point is and I will quote Marco blankets are too strong to account for the way the internal states of an organism are often directly affected by external states without any statistical mediation or loss of Integrity as anyone who's ever been Tippy after a few beers or felt dizzy from heat exhaustion can verify so now we're kind of we've gone to a completely different formulation which is just no vicarious blanket States okay the we're having Direct effects um Jeff I'm going to start with you because I your face is telling me that you don't like that so much um how does that sound to your ears I I'd have no idea I have no idea what is meant by that um it's sort of like I mean and certainly the analogy that was provided I do not understand I I suspect that either I mean I'm probably being obtuse or maybe I haven't I'm not Tippy enough given analogy um but I don't understand how anything can have a direct effect I mean you know I I can think of another scenario that one might appeal to to make a similar argument which is like and I always talk about like protons and electrons they have electric Fields right and those fields have you know an effect right but to my mind that just means you know your blanket is the field right yeah do you have any thoughts on um I was I was imagining that Jeff was not going to like that and and and and I do understand why right I mean it's kind of in in the very definition of marov blanket is those sets of states that are always there kind of you know mediating between the the internal and the external I guess that what Ezekiel was trying to point out there it's something like it is possible for those blankets blankets to be leaky sometimes I think this is an idea that that you find in the in the wig Mar cover blankets um uh paper too uh oh know that that there's no there's no absolute Fusion so to speak between the internal and the and the and the external States uh and that's why they are weak right kind of there's not a full encapsulation ah this is just an an speculation on on on what's a a SLE point there right that those blanket maybe can be leaky in some way sometimes um yeah may maybe this is an issue that I have difficulty understanding because I've already moved on Beyond in my thinking like blankets don't have to be static yeah right because because the the best criticism of of the original set of Mark out blanket stories was your basic

your definition of a blanket or at least the way you are using a blanket prohibits matter transfer between object and environment so like good luck describing something that eats you know right so fortunately you know you know the the way around that originally was to just sort of say oh we could have like made the blanket operate on fluxes instead of particles and I never really liked that you know but but now that we've gotten this issue where we've basically said oh no look Dynamic blankets it all still works right all the math still works out just fine even when there's matter transfer it works on flames and so to my if that was the ultimate issue then I would say that issue has been resolved but I would understand why someone would have drawn that conclusion from the literature because the literature focused on problems that were a lot more solvable like the dynamic so you know identifying Dynamic markof blankets is a really mathematically challenging problem yeah well like whereas specifying a static one makes your life really easy yeah and I you know you know you should never be overly critical of people in my view for you know simple mathemat you know mathematically mathematical simplifications right you should always be generous and say like oh I understand that's not part of the principle per se that's just the particular use of it you know my my favorite example of this is is like general relativity you know the the equation of general relativity has like three terms it's like big G little G and T what they don't tell you is that hidden behind those three terms is a ton of math which is why there's only like one solution to that problem right and the same sort of goes for the free energy principle right if you actually write down what the objective function for active inference is it's really hard to compute yeah and so so they made a lot of assumptions and to simplify the math and to make the computation easier right and so you should always be I always encourage people to be generous with devaluating things in which there's there are equations and then there's simplification you got to if you're evaluating the theory you got to evaluate the original equations right and not criticize the simplifications very christmy um maybe this will help or uh or not or maybe it might muddy The Waters

Another quote now from The Mark of blanket trick um without the mark of blanket formalism the Dual information geometry would be reduced to its intrinsic component which is the only one needed for standard thermodynamics and I've written next to that on my notes why don't we just use this Jeff why don't we just use this because we want I I really do think because like at the heart of a lot of this is like phenotyping it's sort of like the goal is actually to segment the universe into little understand into smaller understandable pieces right and since if that's the goal that's that's sort of you know I sort of I feel like I addressed this before when I said like you know if we could model the whole universe like no problem like why would we

bother to chop it up in the little pieces and sort of say oh no this is a car I'm gonna summarize car Dynamics I'm gonna worry about the little bits and part bits and pieces inside because I could you know why would you do that if you could just simulate the whole thing and it was no problem right so there's sort of this there's sort of an empirical question here like that's why the reason we divide things up the reason we draw blankets around things is to make our lives is to make our ability to understand the universe more tractable right is to make is to make it easier to get right the reason why you know it's like you could always simulate everything if you could you really can't but like if you could you would but you can't right you you're looking for cognitive shortcuts if you're a cognitive scientist you're looking if you're a scientist you're looking for simplifications of complex systems you're looking for macroscopic variables it's it's so it's sort of like there's an empirical reason why you do that vient have you got anything to out of maybe I should have asked exactly what this claim uh entails what what the reduction to an intrinsic component actually looks like actually I that was uh I wouldn't say a quote but that was verbatim almost from from from the from the Dual geometry papers right so it was more than saying more than focusing in the reduction you could do I think what what the what what the the free energy principal proponents are saying they're like look this is just thermodynamics extended a little bit because we have this this this blanket that they allow us to do this this all thing right so that's what what thing and actually I I'm going to one more time agree with Jeff in many things uh uh and I think that again that was that was kind of the spirit of the Marco blanket trick to try to go ourself through the through other things that you need for for the for the whole building to be built so to speak and of course you need to you need to to choose the the the thing you are interested in you need to make assumptions on okay these are the variables I'm going to be working with and with that I mean uh eological psychologists are pretty much the first people that during the 80s started using dynamical systems theory to to Model Behavior actually there's a there's a there's a 1987 book by Peter kogler and Michael arvi called information natural law and rhythmical Behavior if you go there and read how they gather from statistical physics and and all that stuff you see many similar points than than than you see now in the in in in the free energy principle right but the point being when I say okay we we model or we you know that within within this kind of dynamic cognitive science there's a very famous model the hkb model of couple oscillators in which the the the main variable is relative phase why is it relative phase well it's a it's a available very easy to track and when you track it and you and you model it you have a very cool two attractors for these parameters and you change and and you see how the how the two

attractors go to one so you you can you can express the qualitative changes of of the of the cing of the two oscillator in a very simple way with just one variable and you know and but why you pick that one well because you know it's it it's easy and and and it model it well right so so I mean I obviously agree with Jeff on that the point was yes in the literature these simplification sometimes are not clear and yeah it's good to to I think they and and and I also understand yet they are not clear because the aim was generality right so we want this to apply to everything so when when when the couple oscillators guy come but no but talk about my thing okay to that's right yeah we need to put all these other things us to make it work right so kind of thing yeah and that's one of the reasons why I really got to get this paper out so basically it's it's you know if you another way to think about what you know the mark out blanket story is is that it's it's an integral component not just of systems identification Theory but also like macroscopic physics Discovery right so why did you pick the phase difference it was a convenient variable it encapsulated all of the Dynamics right what if your problem was I just have a complex system and I don't know what and I need to discover what the phase variables should be right and the marov blanket story is one way to do that right it's sort of like I want to segate my complex system into different subsystems and if you have a mark blanket Discovery algorithm right then it sort of does that for you it does macroscopic physics discovery that it identifies that that was the simplest variable to use that makes me wonder out of the bloom so I don't have an answer to this is a pure question that might not have an answer when when people who work with dynamical systems or at least part of them to model modor control right they they they gather a lot of ideas from bursting and from hen and the idea of synergy right like like the brain is able to to control the arm as a Synergy because it's already coordinated uh as a as a as a whole unit there could you put so could you put a marov a marov blanket on a Synergy so to speak I don't know so I think that that so so to some extent and so and this is a I'm I'm going to just sort of say this is a gross generalization of what the Discover of what the blanket Discovery algorithm does but in a in a very real way right it's doing dimensionality reduction right which is basically ident Synergy identification right so why is it that like you know I can model I can just sort of look look at all this complicated stuff I can do well it turns out right there's really only like a very small number of variables I have to track because you know they're all correlated right there are restrictions or they're constrained right my wrist is the same distance for my elbow like and so on and so that's another way of character well that means they're correlated right and Synergy and correlations are very are sort of you know are related and so to some extent what the algorithm actually does or when

you're doing Markov blanket Discovery what you're basically saying is what's correlated what aspects of the dynamical system are sufficiently strongly related that I can reduce them to a single variable right so for example if a rigid body right I can describe this whole thing by two points right right I you know you know I don't have to model every single molecule in my arm right I can reduce it to a single Variable and that's another way of thinking about like again macroscopic physics Discovery um right but I also think that's one of the reasons why like the FP is as it's formulated is strongly tied to just empiricism right it's it's it has a lot to do with just how we do science right we're we're doing like dimensionality reduction and looking for principles that uniformly apply um and and I think that the focus of the literature has been on the uniformly AI thing and now we need to bring it into the real world of empirical inference where it's the dimensionality reduction thing but they are in fact completely compatible amazing uh just again to Loop around to what I had said so that people know where to go the quote that I chose is footnote 12 of the Markov blanket track I told you I read the paper um and I think it pertains to this two complementary ways of describing the density dynamics of internal States one being this intrinsic uh change over time towards uh the attractor State and then the other one is this extrinsic um tracking of external States and I'm not a mathematician but it appears to me that the maths seems equivalent um and we can actually make a kind of formal equivalence between those things which is quite quite beautiful actually it's it's almost as if they're equivalent I'm not a mathematician give me a break um no no no no no sorry that was that was not a criticism that was a terrible joke oh okay f f oh right yes yes very good very good it looks like they're the same um yeah excellent so I will finish with what I promised we'd finish with which is a kind of bigger potentially philosophical question that expands beyond the scope of the free energy Principle as is right now um and this is an issue that actually is really matters to me uh when I first discovered active inference I remember tweeting two things one of which was uh the presence of Markov blankets upon Markov blankets is the best argument from creation that's ever been and the second one was um I think that the apparent distinctions of internal and external States mediated for a Markov blanket might itself be an illusion and there may just be some sort of singular reality I'm paraphrasing my tweets but it was more just philosophical real thinking and we mentioned pan psychism earlier um and again I don't want to get too into the weeds on this but Jeff maybe we could start with you where do you think this all comes from these um these distinctions that we can carve for instrumental reasons which seem to you know which do work mathematically and what are your hunches about what might lie underneath them I love questions like this and I

have the perfect answer so what lies underneath nothing oh that that was my serious face that doesn't help no no so so I remember one time I got asked a question like you know so we're talking about like you know p's and q's because that's what you're doing when you're doing probabilistic reasoning and somebody who raises their hand like is there a true P the answer is no no no there's not and it's not because there isn't a true PE right it's just I don't understand the question right it's sort of a it's sort of a jerky thing to say I I often describe when I when when I get so as you might imagine given what I do for a living I get ask questions like that all the time and I usually refuse to answer and the reason I refuse to answer is because I got up to work today I put on my coat and tie and I put on my work pants right my work pants are my harsh instrumentalist pants and then if you want me to answer like philosophical or religious questions it's like well I have to put on a different pair of pants right you know when you know when functioning in this in this environment like you know it's it it you know I I take a very very instrumentalist view right and I think the you know so you ask like oh do we carve you know is there some is it all just one thing right are these artificial carvings AR the artificial carvings are artificial right we carve up the world because it's convenient right because it simplifies the problems we have to solve it has empiric it has predictive power and empirical utility I like to say that science is not about truth it is about prediction and data compression and nothing else right so when functioning as a scientist that's what I focus on which is my way of saying like that's that's my answer harsh instrumentalism harsh instrumental means you don't have to worry about like what's true I can't wait for all the philosophers to be shouting at you virtually um viene what are your thoughts on this I was expecting this answer from Jeff because my preparation to this podcast was not to reread my own paper but to to watch another podcast on the machine learning Street talk I think and yeah that he was saying electron is just the name for an equation you know like right so so I was expecting heart's instrumentalism I kind I'm kind of an I'm I'm I'm kind of an instrumentalist too in some ways uh but at the same time I'm a realist so how can that be how can that be the case right I I guess it's it it comes overall um comes all to to a form of pragmatism in the end in which you can be realist about the kind of things you do uh and if they are successful well you you you are dealing with the with the with the real world and um you do so by using instruments and I I'm in favor of aurality of instruments uh and marov blankets are one of them that I I don't personally use in my own research right but you can use it um but I wouldn't try to find a final Foundation like oh there's only one um or or you know or everything is such or everything is such other thing right it's more like uh we act around um

and reality is somehow the way all where all our actions converge um and not actually sometimes do sometimes don't and and we we can reality in that way and all we have is our our Concepts that are instruments and are you know not foundational in any way tools marov blankets other tools and we go like that but I mean I don't think there's that there's a there there there's some form of you know matter of fact or fact of of of of the matter so uh so speak so there's something that that there's something we some reality that we build up together through action and is there it's just it can change so it's not the final thing it's it's it's it's not the an old philosopher you know an ancient philosopher would say something like the the AR right this this this final original thing there's no but there's a thing we do there's there's a there's there's a matter of fact and and theep variety of tools and their superposition and their obus OB obus I don't know that that that word in English but even the even the things that they don't match well tell us thing about what what kind of thing we can do and we can do oh where are my Buddhist friends where are I where my well so that's that's the funny you asked the wrong question if you wanted me to put my put on my Quaker Buddhist pants I can do that too oh I just generally refuse to maybe on Christmas day I'll ask it's one of those it's like I I remember one of my uh my P one of my uh one of the faculty in my department when I was getting my PhD once said that um uh philosophy is a lot like like algebra or or or going to the bathroom it's just not something you should do in public I don't agree with that you know it it indicates like why it is I hesitate to answer these questions because I don't really have good answers I have my answers and Def faulting to instrumentalism is like just the easy way out right sure well yeah they're purposefully quite provocative and probing uh but people seem to like answering them chaps that was uh that was tremendously fun um maybe we could have just a couple minutes for closing remarks um given we've reached such a beautiful point of conciliation viene um maybe in addition to sort of closing remarks you could also let people know what you're working on and where they could reach out to you if they wanted to know more about your topics of Interest okay so I'm going to start by by talking one one last time about the marov blanket trick the paper kind of grew on its own it's way long I think it's 24,000 words so it's it's way it's way longer than I was expecting uh uh because actually most most of it is just uh me asking a qu not me we all of us right but asking the questions we have right oh but if if the formalism work why this and why that and why candle flames and why you know that kind of thing but I really don't have an strong uh position on those things I I just would like to know okay how would you apply this to a model with a relative face and that that seems to include the whole system I only have a strong opinion with one thing and which

is uh the fact that for perception you need or not need prior knowledge right that's that's that's actually the real my my real State there like I think uh every time you endorse the system with some prior knowledge in order to make sense to go as as we talk from the from the impoverished impoverished input to the real world so to speak uh you have no way to justify that that that knowledge or to explain what's why it is there right the typical way you know what say no it's Evolution right but it's kind of unsatisfactory for many I don't want to go to this thing right but this is actually my only kind of my my Ideal breaker I think theories of perception should then appeal to PRI your knowledge to explain it and I think one theory that that does that is ecological Psychology by virtue of the lawful relationship between the patterns of light and the environment you have in those patterns of light information enough as to know exactly what the exactly to to to to to a good the degree what the what the real environment is with without the need of pulling other kind of knowledge into the equation and I work on that right so so so so part of my work is is is trying to make sense of a cognitive Neuroscience that starts with ecological psychology for instance if the brain is not in the business of inferring the real causes of impoverished stimulation or of enriching impoverish stimulation with other knowledge as to build a model of the environment what the hell is the is is the brain doing right so within the ecological approach the word uses resonance the brain resonates to this richer information and part of my theoretical work is to try to operationalize the resonance metaphors so to speak so how can we have a a concept that we can actually test and part of my empirical work I'm kind of maybe 70% a philosopher 30% a cognitive scientist cognitive neuroscientist is to try to test that so this year I've been in in in in the bmobile lab which is at the technical University of of Berlin and they and they specialize in Mobile EG and so I've been Gathering some multimodal data that that that has you know ey tracking motion capture and EG to try to find what kind of patterns of information our brain can be resonating to so how the brain Dynamics can be constrained by this specific structures and not just by by by kind of say simple input and that kind of thing so I'm trying to yeah last sentence I'm trying to do something like an ecological Neuroscience uh both in the theoretical realm and the and in the experimental uh realm very cool very cool Jeff go ahead uh so I'm taking the exact opposite approach I'm trying to see how far we can push this whole prior thing I I like to joke that the the difference between a basian and a non- basian is honesty right is you have priors right I don't know where they come from I don't know that they're normative but they're there you you're kind of stuck with them all knowledge is like llas all knowledge is based on um all knowledge is probabilistic right it's always based on assumptions there are

two sets of assumptions that are critical to worry about one is I'm going to get to what I'm doing in a second one is the data right any conclusions you draw is conditioned on your observations but there's another important thing that we often forget about right is that all the conclusions draw are also conditioned on your models for the universe and I say models this is just my view I could see vente is going oh my God no um when I say models I actually think the most critical thing that we often neglect when we talk about the basian brain is that there's not one model there's a lot of them and they and and they and and you don't just use one at a time right you use lots of in fact you one you could even think and this is very coarse I'm I literally am gonna get to what I'm doing in a second it's very course but you could think of a basian brain as having brain regions right and don't overp parse this because I know people will right and each region is a model right each region describes is a sta statistical model of something and when you get a new task what do you do right well you have to figure out which collection of models and how they're connected right are the is you know are the ones that like solve your problem or model or model the data or describe the data or compactly summarize this like the data compression compactly summarize the information that you received or give you the predictions about the things you care about or however you want to phrase it and so what we're trying to do right now um is we're exploring sorry we're exploring the extent to which we can get this to work at you know at scale so we want to construct a brain that's not like some fast feed forward neural network right but one that is more brain like right by summarizing what by saying that no no no brain has lots of there's lots and lots and lots of models and the way that the brain actually attacks problems is by reasoning by analogy this model explains this part of what's going on this model explains this part of what's going going on and the trick is how do you figure out how to connect them and so this is a massive model comparison SL basian graph neural networking type problem which is like computationally intensive like that's a i' I'd put that in quotes like but it would like it would be like several sets that's that's how much scare it is um that's what we're working on right we're working on pushing you know seeing how far we can really push this basum brain hypothesis but not just sort of saying there's a prior by saying there's lots of them and they interact in interest way and at that point is it still a prior viente or has it become something else right I mean uh this is kind of the of the agreement I I I I read with with M well for instance but with other people there right yeah I mean and I think it is good I I don't like the the uh the idea of of prior knowledge or one prior or many of them I think there are are other ways we like to talk about ecological constraints that are not inside here out there well but blah blah blah blah

blah but what I try try try to do is to is no priors let's see how far I can go and it's fa that you guys do okay this this set of of of priors let's see how far and that's it that's I think that's good for science actually yeah no that's the whole point right because eventually what we want to do is basian model comparison between your approach and mine because you know basian is the only way to do it obvious I think you guys are gonna absolute pleasure absolute pleasure chatting with you well it's no it's been an absolute pleasure to witness it um thank you guys for coming on the podcast and uh I'm sure there are some people out there devilish people who expected pots and pounds to be thrown across webcams but um it I you know this is what science is about at the end of the day um finding convergent paths and trying to get closer to some approximation of Truth so I hopefully we've contributed sorry no and you both Lean Back simultaneously when I said that word um okay some something that helps us predict and data comp there you go Jeff um but thank you guys again from me and The Institute and uh uh yeah everyone listening definitely check out Jeff and vent's work so thank you again thank you